[INCH-POUND]
A-A-58042
May 3, 1995

COMMERCIAL ITEM DESCRIPTION

ANALYZER, DIGITAL DATA, ELECTRONIC TEST

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

This CID describes a Digital Data Analyzer with a minimum of 32 channels used for collecting data acquired in the timing and state analysis modes. This CID outlines the minimum requirements for a logic analyzer which operates in the 25 MHz to 100 MHz range. Only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. CLASSIFICATION.

Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description.

3. SALIENT CHARACTERISTICS.

3.1 Design and Construction. The equipment shall be solid state and enclosed to provide protection from mechanical shock and conditions associated with bench-top, rack-mount and occasional portable use in widely diverse, environmentally controlled locations. The equipment shall be UL (Underwriter's Laboratories) listed and approved and/or shall comply with the Type III safety requirements of MIL-T-28800.

3.2 Electrical Power Sources. The equipment shall operate from nominal commercial, military and shipboard power sources of 110/220 volts AC (+-10%) at line frequencies of 50 Hz to 60 Hz.

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any data which may improve this document should be sent to: The Resources and Logistics Services Division, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB TX 78241-6524.

AMSC N/A FSC 6625

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3.3 Temperature Requirements.

(a) Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to plus 50°C.

(b) Non-operating Temperature. The equipment shall meet its performance and accuracy requirements in a non-operating environment of minus 40°C to plus 60°C.

3.4 Pre-conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. Any calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

3.5.1 Calibration Interval. After calibration, the equipment shall meet each performance requirement within the tolerance specified for a period of at least 9 months based on an average operating time of 1500 hours.

3.6 Performance Characteristics. Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance characteristics shall apply, with or without probes, in a laboratory ambient operating temperature.

3.7 Display. A CRT or other video display having minimum useful scan of 127 mm (5 inches) wide by at least 102 mm (4 inches) high shall be provided. The display must provide sufficient luminance for viewing under varied light conditions. CRT phosphor shall have a high resistance to burn-in. The worldwide phosphor-type designation system (WPDS) GH phosphor shall be the reference for this requirement (formerly phosphor type P-31).

3.7.1 Full-Screen Capability. The display shall have the capability to display at least 24 lines of 32 characters or bits.

3.7.2 Data Display Formats. The logic analyzer shall be capable of displaying asynchronously or synchronously sampled data in timing or state format. State format shall be selectable from binary, octal, hexadecimal, and ASCII. The unit shall provide an indicator to show activity at each test point.

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3.7.3 Display Magnification. The analyzer shall be capable of displaying data in the waveform format to check signal timing and measure time differentials with at least a times one (x 1) and a times ten (x 10) horizontal magnification of at least six simultaneous channels.

3.7.4 State Mode Display Capability. The analyzer shall be capable of displaying in a numeric format a minimum of 15 consecutive samples of all data channels which sample data synchronously. Full memory depth shall be displayed through paging, scrolling, or similar techniques.

3.7.5 Trigger Word and Cursor Word Display Enhancement. The position of the trigger word relative to the sampled data set, and the currently displayed data within the sampled data set, shall be readily discernible by such means as: highlighting, numeric indicators, or position indicators such as windowing and cursors.

3.8 Inputs. All data inputs shall be sampled asynchronously at 100 Mega samples per send and synchronously at up to 25 Mega samples per second.

3.8.1 Data Channels. A minimum of 32 data channel inputs shall be provided. Glitch capture shall be available on one half of the channels.

3.8.2 Input Impedance. Input impedance shall be at least 100 Kohms at plus or minus 2% paralleled by no more than 15 pf as measured at the probe (test lead) tip.

3.8.3 Maximum Non-destructive Input Voltage. The equipment shall withstand at the probe tip, voltages not to exceed plus or minus 40 volts (DC plus peak AC). Cross channel differential voltage is not to exceed 80 volts (DC plus peak AC).

3.8.4 Logic Threshold Selection. Logic threshold levels of the inputs to the high impedance probes shall be selectable from a fixed TTL threshold or variable from minus 3 volts DC to plus 6 volts DC in 100 millivolt increments or smaller. The equipment shall have provisions for monitoring the variable threshold settings and displaying the current value of the variable threshold. The equipment shall be capable of assigning different thresholds to groups of data channels so that dual logic level settings may be monitored at the same time.

3.8.4.1. Accuracy. The equipment shall have a logic threshold accuracy greater than plus or minus 5% of threshold and plus or minus 5% of total variable voltage range.

3.9 Clocks. The equipment shall be capable of operating from an internal (async) or an external (sync) clock.

3.9.1 Clock Qualifiers. The equipment shall contain at least two clock qualifier inputs which can be logically combined to qualify the acquisition of data in synchronous operation.

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3.9.2 Selectable Data Storage. The equipment shall have the ability to inhibit or enable the storage of incoming data dependent upon that matching a previously specified data qualifier. Each channel of the data qualifier shall be individually selectable as active high or active low.

3.9.3 External Clock Active Transition Selection. The equipment shall permit the operator to select either the positive or negative or both external clock transitions as the active transition.

3.9.4 External Clock. The equipment shall be capable of sampling synchronously with an external clock through a high impedance clock input. The clock input period shall be variable over a minimum range of 20 nanoseconds to 1 second.

3.9.5 Internal Clock. The interal clock rate shall be adjustable from 10 nanoseconds to 20 milliseconds if transitional timing is not used.

3.10 Memory. The logic analyzer shall have a minimum memory depth of 1K bits per channel across all channels at full acquisition speed.

3.11 Sampling.

3.11.1 Asynchronous Sampling. The asynchronous sampling rate accuracy shall be plus or minus .01% and the channel skew shall not exceed 4.0 nanoseconds typically.

3.11.2 Synchronous Sampling. The analyzer shall be capable of operating with a minimum clock pulse width of 10 nanoseconds. The minimum data set-up time shall be no greater than 12 nanoseconds before the active clock transition. The minimum data hold time shall be no greater than 2 nanoseconds after the active clock transition.

3.11.3 Glitch Capture. The analyzer shall be capable of capturing glitches as narrow as 7.5 nanoseconds with 50 millivolts of input or 25% of the total voltage swing, whichever is greater. A glitch shall consist of more than one transition of the input signal within the asynchronous clock interval.

3.11.3.1 Glitch Memory. With the glitch detector enabled, the associated dynamic or reference memory will be at least one-half the data channels wide by at least 512 bits deep.

3.11.3.2 Glitch Display. Glitches will be uniquely identified. Procedures such as inverse video, highlighting, or the use of unique characters to represent glitches shall be acceptable.

3.12 Triggering Modes.

3.12.1 External Trigger Input. The equipment shall provide a trigger input which shall enable triggering with a TTL level input.

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3.12.2 External Trigger Output. The equipment shall provide a trigger output at the TTL level which signals the occurrence of triggering.

3.12.3 Trigger Word Recognizer Inputs. The equipment shall have all data channels available as inputs to the trigger word recognizer. All channels of the trigger word recognizer shall be individually selectable as "active high", "active low", or "don't care". Active-on glitch capability shall be provided for all channels.

3.12.3.1 Filter. An adjustable filter must be provided which indicates how long the word recognizer must be true or false in order to be a valid occurrence during asynchronous operation. The minimum filter time shall be 20 nanoseconds for asynchronous mode.

3.12.3.2 Trigger Delay. A provision to delay recognition of the trigger word in the asynchronous mode by one of several discrete time intervals selectable by the operator. The equipment shall be capable of delaying triggering up to 5000 clock samples after the occurrence of the last trigger level.

3.12.3.3 Trigger Word Combinatorial Selection. All trigger word inputs and word qualifier inputs shall be individually selectable as "active high", "active low", or "don't care".

3.12.4 Sequence Triggering. The equipment shall be capable of accepting a sequence of at least eight trigger levels. Each level shall include the occurrence of the word recognizer a specific number of times up to a maximum of 1,000 specifiable number of occurrences.

3.12.5 Restart. The equipment shall have the ability to restart the trigger sequencer back to the first level upon the occurrence of an independently defined word or event.

3.12.6 Trigger Position. The equipment shall provide the operator a selection of the trigger point within the data storage window. The selection shall allow from 0 to 100% of the acquired data to be displayed after the trigger.

3.13 Timing and State Relationships. The logic analyzer shall be capable of being operated in four submodes in which the timing channels are operating asynchronously and the state channel synchronously.

The submodes may be:

1) State simultaneously triggers timing; or
2) timing simultaneously triggers state;
3) State simultaneously arms timing; or
4) timing simultaneously arms state;

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3.14 Accessories. The equipment shall be provided with at least, but not limited to, the following accessories:

1) Complete set of probes, leads, and chip clips with the ability to probe standard and surface mount components including plastic-leaded chip carriers of 20 to 68 pins and small outline integrated circuits of 8 to 24 pins.
2) power cord;
3) operation and service manuals;
4) spare fuse if needed;
5) training package in the form of manuals, video, and/or software.

4. REGULATORY REQUIREMENTS.

In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. PACKAGING.

The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specified in the contract or order.

6. QUALITY ASSURANCE.

Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices.

The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract.

The Contracting Officer has the option of accepting or rejecting the product.

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7. NOTES.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

Federal and Military specifications are available from:

Defense Documents Order Desk
700 Robbins Avenue, Bldg. 4D
Philadelphia, PA 19111-5094.
Phone: (215) 697-2179

Known Acceptable Products. The following product(s) are known to have previously met the salient characteristics of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required:

Manufacturer	Model	National Stock Number
John Fluke Manufacturing Co.	3580-30	NSL
Hewlett-Packard Co.	1663A	6625-01-387-0613

MILITARY INTERESTS:

PREPARING ACTIVITY:

Air Force - 82

Custodian:
Air Force - 99

Reviewing Activities:
Army - ER
Navy - AS
DLA - ES

Project No. 6625-0807

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